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Presenter

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**Design and Development of a Flexible AI System for
Financial Market Analysis:**

From Basic to Hybrid and Multimodal Approaches

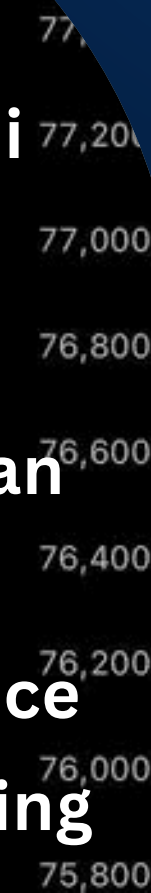
FinScope

Diagnostic Evaluation of Qualifier-to-Value Binding in Financial QA

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Fact. Computer Engineering**





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OVERVIEW

01

Financial AI

02

Systematic
Review

03

Financial
Understanding

04

Financial QA

05

Qualifier to Value
Binding

06

FinScope



PREDICTION VS UNDERSTANDING

- Forecasting prices, returns, or market trends
- Primarily based on numerical and time-series data
- Common approaches include LSTM, GRU, and Transformer-based models

- Main question:
What will happen next?

Historical Market Data →
AI Model →
Future Price / Return

- Interpreting financial reports and documents
- Extracting relevant facts and answering financial questions
- Reasoning over text, tables, and numerical information

- Main question:
What does the financial information mean?

Financial Document →
AI Model →
Answer / Evidence / Reasoning





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02

SYSTEMATIC REVIEW

01

Surveys on
Numerical Time-Series and Deep Learning Foundations

02

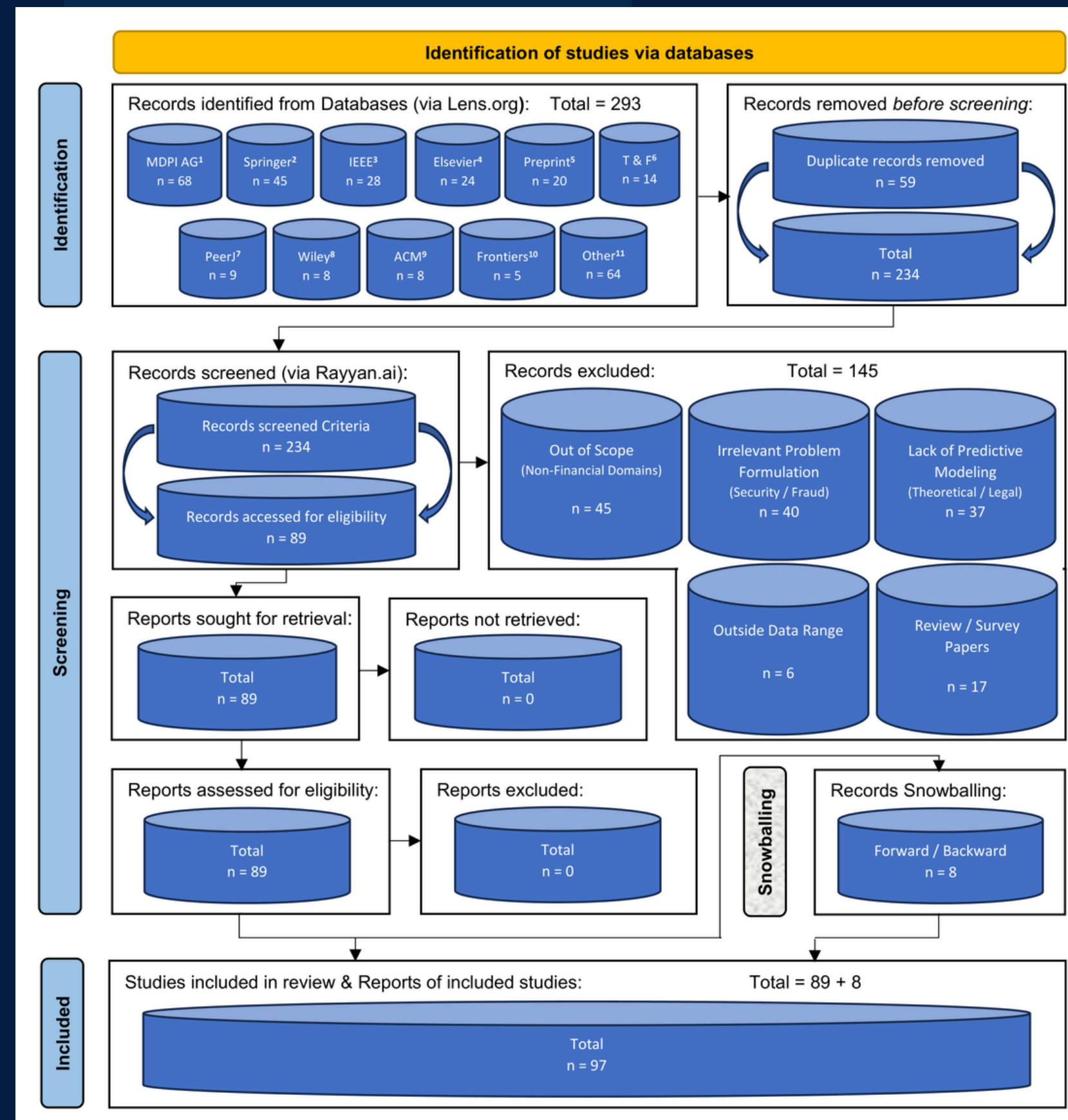
Surveys on
Textual Analytics, Sentiment, and LLMs

03

Surveys on
Visual Analytics and Pattern Recognition

04

Surveys on
Hybrid and Multimodal Approaches





03 FINANCIAL UNDERSTANDING

From Review to Financial Understanding

- Visual-only: 54 studies
- Numerical-only: 23 studies
- Multimodal visual + text: 20 studies
- Strong emphasis on visual and numerical approaches
- Growing interest in multimodal methods
- Comparatively less emphasis on text-centered financial understanding



Can AI systems correctly understand and reason over financial information?



03 FINANCIAL UNDERSTANDING

Why Financial Understanding Matters

- Financial documents often contain multiple valid values for the same metric.
- The meaning of a value depends on its context, reporting basis, period, scope, and qualifier.
- Finding a number in the document is not enough.
- A reliable AI system must understand which value belongs to the requested context.



Correct retrieval does not guarantee correct understanding.

A broad error category may be sufficient for scoring, but not for understanding or fixing model behavior.



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04

FINANCIAL QA



UNDERSTAND THE DOCUMENT

Interpret financial text, tables, and reported values



UNDERSTAND THE QUESTION

Identify exactly what financial information is being requested



SELECT THE RELEVANT EVIDENCE

Locate the evidence that matches the requested context

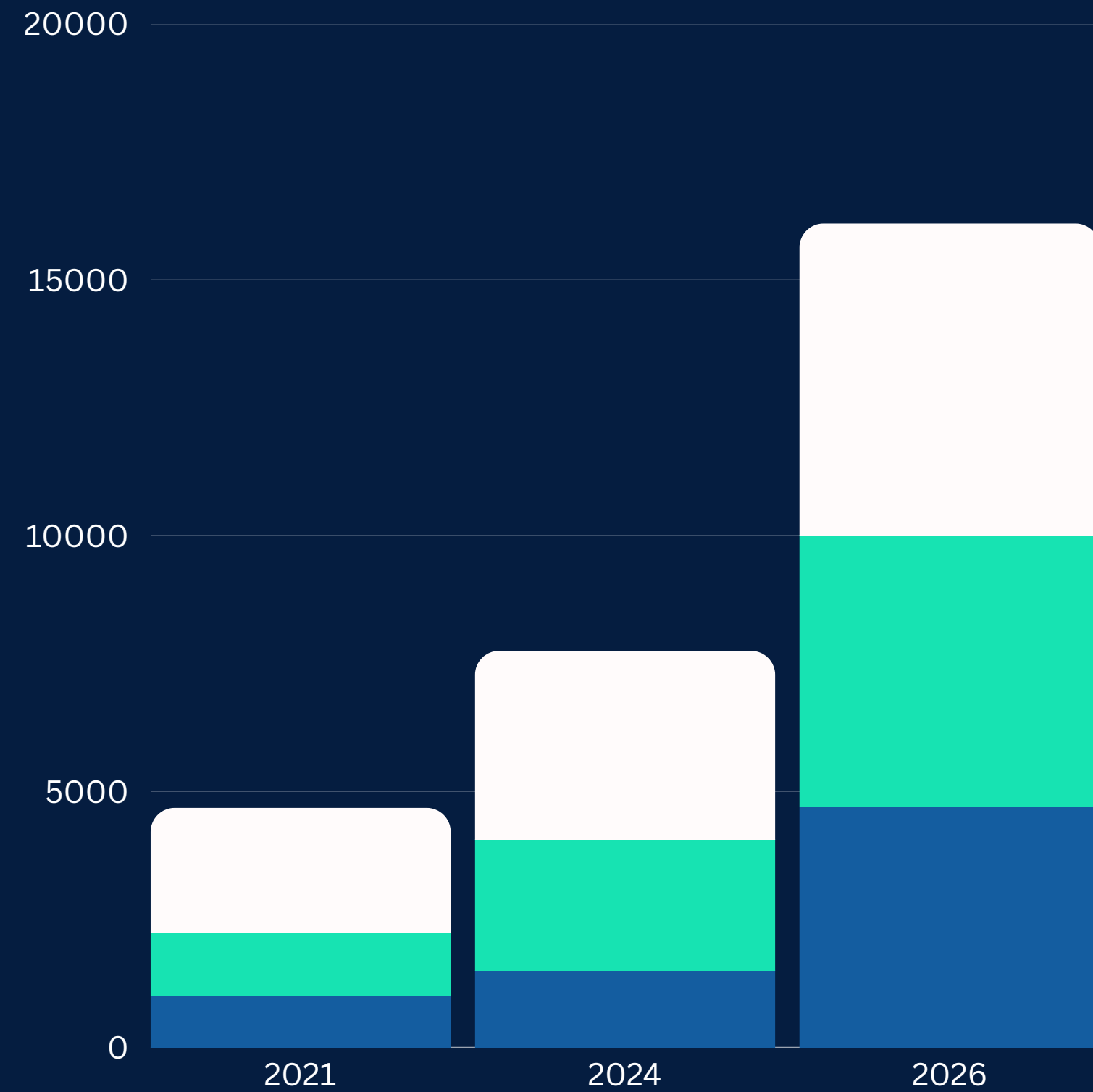


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05

QUALIFIER TO VALUE BINDING

- Presentation Basis
Reported vs. Adjusted
- FX Basis
Reported Currency vs. Constant Currency
- Guidance Position
Lower / Midpoint / Upper Guidance
- Reporting Scope
Consolidated vs. Segment / Region





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Real-World Sample
CONFIG_ID: FSRWFINAL_CFG_004
Source Dataset: TAT-QA
Canonical Axis: FX_BASIS

“Within Cloud & Cognitive Software, cloud revenue of \$4.2 billion grew 40 percent as reported and 42 percent adjusted for currency year to year, reflecting the acquisition of Red Hat and client adoption of our hybrid cloud offerings.”

Example

- reported → 40%
- constant currency → 42%



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Example

Twin A

QUESTION_ID: FSRWFINAL_CFG_004_A

Question:

What was the year-to-year growth in cloud revenue in 2019 as reported?

Target Qualifier: reported

Gold Value: 40%

Competing Qualifier: constant currency

Competing Value: 42%

Twin B

QUESTION_ID: FSRWFINAL_CFG_004_B

Question:

What was the year-to-year growth in cloud revenue in 2019 adjusted for currency?

Target Qualifier: constant currency

Gold Value: 42%

Competing Qualifier: reported

Competing Value: 40%



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Example

Wrong Qualifier Source Value (WQSV)

A source-present value associated with a competing qualifier is selected instead of the value associated with the requested qualifier.



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FINSCOPE

06



REAL-WORLD ERROR INSPECTION

Observed recurring cases where models selected a valid but contextually wrong value.



QUALIFIER IDENTIFICATION

Recurring qualifier patterns were identified from the observed cases.



CONTROLLED DATA GENERATION

Synthetic configurations were created to isolate qualifier-to-value binding.



REAL-WORLD DATA CONSTRUCTION

Real-world Financial QA examples were extracted and reformulated into the same evaluation structure.



VALIDATION & QUALITY CONTROL

Examples were validated, repaired when necessary, and checked for ambiguity and shortcuts.



MODEL EVALUATION

Models were evaluated using accuracy and diagnostic error labels such as WQSV.



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FINSCOPE DATASET COMPOSITION

SYNTHETIC DATA

720 Configurations — 1,440 Questions

- V1: 160 configurations / 320 questions
- V2: 320 configurations / 640 questions
- V3: 240 configurations / 480 questions

REAL-WORLD DATA

136 Configurations — 272 Questions

- FinQA
- TAT-QA
- FinanceBench



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VALIDATION & EVALUATION PIPELINE



1. LLM-Assisted Review
2. Human Verification
3. Adjudication & Repair



FinScope Questions
→ Model Response
→ Normalizer
→ Scorer
→ Diagnostic Label

- CORRECT
- WQSV
- OTHER_WRONG
- INVALID

1. Exact duplicate check
2. Near-duplicate check
3. Evidence overlap check
4. Prompt duplication check
5. Shortcut / leakage checks



RESULT

FinScope Track	Gemma-2-2B-IT	GPT-OSS-120B	GPT-5.6 Sol
V1	37 / 81.25%	29 / 89.38%	14 / 95.63%
V2	38 / 71.56%	2 / 96.56%	0 / 100.00%
V3	68 / 85.42%	14 / 97.08%	6 / 98.75%
Real-World	22 / 59.93%	0 / 90.44%	0 / 91.54%



Key Findings & Conclusion

What the Results Show

- Higher overall accuracy generally coincides with fewer WQSV errors.
- However, high accuracy does not fully eliminate qualifier-binding failures.
- A value can be present in the source and still be the wrong answer.

Why This Matters

- Accuracy alone does not explain how a model fails.
- FinScope provides a diagnostic view of qualifier-sensitive errors.
- It separates semantic binding failures from broader wrong-answer categories.

Main Contribution

FinScope provides a controlled benchmark for evaluating Qualifier-to-Value Binding in both synthetic and real-world Financial QA.

Future Direction

Qualifier-aware prompting · Verification · Contrastive/Twin Training



- 328 configurations reviewed
 - 136 Real-World
 - 192 Synthetic
- 290 initially VALID
- 38 UNCERTAIN
- Adjudication:
 - 21 Confirmed Valid
 - 17 Repair Required
 - 0 Dropped
- 17 / 17 repaired items passed re-validation
- Final benchmark:
 - 856 configurations
 - 1,712 questions
 - FINAL_FROZEN

 Sayed-Hossein-Hosseini execution: add final seven-model run launchers 033ed5b · 5 days ago 24 Commits		
 configs	chore: create complete FinScope repository skeleton	2 weeks ago
 data	data: add FinScope v1.1.1 final frozen release	last week
 docs	evaluation: freeze FinScope evaluation protocol v1.0	last week
 evaluation	evaluation: close pre-rerun readiness and fix Gemma serializa...	last week
 provenance	FinScope provenance Update	2 weeks ago
 results	evaluation: close pre-rerun readiness and fix Gemma serializa...	last week
 scripts	execution: add final seven-model run launchers	5 days ago
 src	chore: create complete FinScope repository skeleton	2 weeks ago
 tests	chore: create complete FinScope repository skeleton	2 weeks ago
 validation	evaluation: close pre-rerun readiness and fix Gemma serializa...	last week
 .gitattributes	chore: initialize FinScope repository structure	2 weeks ago
 .gitignore	chore: initialize FinScope repository structure	2 weeks ago
 CHANGELOG.md	chore: initialize FinScope repository structure	2 weeks ago
 CITATION.cff	chore: initialize FinScope repository structure	2 weeks ago
 LICENSE	chore: initialize FinScope repository structure	2 weeks ago



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THANK YOU



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**THANK YOU FOR
YOUR ATTENTION**

DEDICATED TO MY DEAR PARENTS

